0. **Code:** sotomayorj1594 Sotomayorj1594.zip

1. B. Sentinel Control Loop/Program.cs
2. C. Min, Max/Program.cs
3. Test01 A.Selection statements/Program.cs

1. **B. Sentinel Control Loop/Program.cs**

```
using System.Runtime.Intrinsics.X86;

namespace B._Sentinel_Control_Loop
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int num1 = 1;
            int num2 = 1;
            int count = 0;
            int sum = 0;
            while (num1 ≠ 0 && num2 ≠ 0)
            {
                Console.WriteLine("Enter a pair of integers. ");
                num1 = int.Parse(Console.ReadLine());
                num2 = int.Parse(Console.ReadLine());
                if (num1 ≠ 0 && num2 ≠ 0)
                {
                    int quotient1 = num1 / num2;
                    int quotient2 = num2 / num1;
                    if (quotient1 ≠ 0)
                    {
                        int remainder1 = num1 % num2;
                        if (remainder1 ≠ 0)
                        {
                            sum = sum + remainder1;
                            count++;
                        }
                    }
                    else if (quotient2 ≠ 0)
```

B1

B2

B3

B2

will not calculate if num1/num2≠0.

1. **B. Sentinel Control Loop/Program.cs**

```
                    {
                        int remainder2 = num2 % num1;
                        if (remainder2 != 0)
                        {
                            sum = sum + remainder2;
                            count++;
                        }
                    }

                }
                else
                {
                    break;
                }
            }
            if (count > 0)
            {
                double avg = (double) sum / count;
                Console.WriteLine($"The average is {sum}/{count} = {avg}");
            }
            else
            {
                Console.WriteLine("The average cannot be calculated");
            }
        }
    }
}
```

B3

B1

B4

2. C. Min, Max/Program.cs

```
namespace C._Min__Max
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter integers until they are identical");
            int maxPrev = int.MinValue;
            int MinPrev = int.MaxValue;
            int num = 0;
            int prev = 1;
            int count = 0;
            while (prev ≠ num)
            {
                prev = int.Parse(Console.ReadLine());
                if (num % 10 > maxPrev % 10)
                {
                    maxPrev = num;
                    if (num % 10 < MinPrev % 10)
                    {
                        MinPrev = num;
                        count++;
                    }
                }
                num = int.Parse(Console.ReadLine());
            }
            Console.WriteLine("The count is " + count);
        }
    }
}
```

min_count = 0;
max_count = 0;

num

C2 X

if (count > 0){

missing

C3

prev

if (minprev < num){
min_count = count;}

X

prev = num;

ask twice in each iteration!

C4

or one

max_count - min_count - 1

2. C. Min, Max/Program.cs

```
}
```

3. Test01 A.Selection statements/Program.cs

```
namespace Test01_A.Selection_statements
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter number 1");
            int num1 = int.Parse(Console.ReadLine());
            Console.WriteLine("Enter number 2");
            int num2 = int.Parse(Console.ReadLine());
            Console.WriteLine("Enter number 3");
            int num3 = int.Parse(Console.ReadLine());
            int count = 0;
            double sum = 0;

            if (num1 ≠ 0 && num2 ≠ 0)
            {
                int quotient1 = num1 / num2;
                int quotient2 = num2 / num1;
                if (quotient1 ≠ 0)
                {
                    int remainder1 = num1 % num2;
                    if (remainder1 ≠ 0)
                    {
                        sum = sum + remainder1;
                        count++;
                    }
                }
                else if (quotient2 ≠ 0)
                {
                    int remainder2 = num2 % num1;
```

A1

A2

Logic error:
num1 = 0
but
num2 ≠ 0
is feasible.

go here

A3

go here

A3

if(num1 ≠ 0){

3. **Test01 A.Selection statements/Program.cs**

```
                if (remainder2 ≠ 0)
                {
                    sum = sum + remainder2;
                    count++;
                }
            }
        }
        if (num2 ≠ 0 && num3 ≠ 0)
        {
            int quotient3 = num2 / num3;
            int quotient4 = num3 / num2;
            if (quotient3 ≠ 0)
            {
                int remainder3 = num2 % num3;
                if (remainder3 ≠ 0)
                {
                    sum = sum + remainder3;
                    count++;
                }
            }
            else if (quotient4 ≠ 0)
            {
                int remainder4 = num3 % num2;
                if (remainder4 ≠ 0)
                {
                    sum = sum + remainder4;
                    count++;
                }
            }
        }
        if (num3 ≠ 0 && num1 ≠ 0)
```

A3

A2

A3

if (num2 ≠ 0){

A3

Same

3. Test01 A.Selection statements/Program.cs

```
        {
            int quotient5 = num3 / num1;
            int quotient6 = num1 / num3;
            if (quotient5 ≠ 0)
            {
                int remainder5 = num3 % num1;
                if (remainder5 ≠ 0)
                {
                    sum = sum + remainder5;
                    count++;
                }
            }
            else if (quotient6 ≠ 0)
            {
                int remainder6 = num1 % num3;
                if (remainder6 ≠ 0)
                {
                    sum = sum + remainder6;
                    count++;
                }
            }
        }

        if (sum > 0 && count > 0)
        {
            double avg = (double) sum / count;
            Console.WriteLine($"The average is {sum}/3 = {avg}");
        }
        else
```

A2

if(num3 ≠ 0){

A3

not needed because Sum < 0 is a valid average

A4

3. Test01 A.Selection statements/Program.cs

```
            {
                Console.WriteLine("The average cannot be calculated");
            }

        }
    }
}
```

A4